

QuantROI System Architecture

High-Level Architecture Overview

QuantROI RIA Roboadvisor Platform

Frontend Layer

- React UI
- Voice Interface
- Mobile App
- (Trading)
- (Grok/OpenAI)
- (React Native)

API Gateway Layer

- FastAPI Enterprise APIs
- € ZKP Authentication
- € Oracle Performance
- € Compliance
- € Vote Submission
- € Causal Analysis
- € Voice Co

Core Business Logic

- System
- ZKP Voting
- Causal AI
- Orchestrator
- Pipeline
- Engine
- € Delegation
- € Groth16 ZKP
- € CausalN
- € Coordination
- € RL Vote IDs
- € DoWhy
- € Oracle Mgmt
- € Delay Detect
- € NVFP4 O

Oracle Optimization Layer

Oracle Management System

- Supra
- OR
- Audited
- Chainlink
- Oracles
- Libraries
- VRF
- € Zero-blk
- € Verified
- € Verifi
- € <1s feeds
- € Compliant
- € Random
- € <100ms
- € ZK-Rollup
- € Batch
- € TTL Mgmt
- € Local Fin
- € Optimi

Blockchain Layer

Solana Smart Contracts

Delegation Management Payment System RIA Compliance

Founder Auth Auto-payment Audit Trail

Duty Record Milestone Pay SE

Oracle Batch Contract Terms Co

Knowledge ZKP Strategy Verification AI Auditor Proof Verify Completeness Strategy Val Sincerity Risk Mgmt

Data Layer

Neo4j Redis IPFS

Graph Database Cache Distributed Storage

Causal Nodes Oracle Cache Immutability

Vote Nodes Query Cache Hash St

News Nodes Session Data Audit T

Relationships <100ms Access ZKP Pro

Infrastructure Layer

Kubernetes Cluster

ArgoCD Prometheus Datadog

GitOps Monitoring Observ.

MLflow Kafka Load Bal.

Hierarchy Streaming Nginx

Component Architecture Details

1. Oracle Optimization Architecture

Oracle Optimization System

Oracle Provider Layer

MUTUALLY EXCLUSIVE OPTIONS (Either/Or, Not Both)

Option A: Supra Oracles Option B: Audited Libraries

â€¢ Zero-block delay â€¢ SEC-compliance

â€¢ <1s data feeds â€¢ Verified inputs

â€¢ High-frequency updates â€¢ Audit trails

â€¢ Consensus pricing â€¢ Enterprise-ready

Enhancement Components

Chainlink VRF Redis Cache Polygon Miden

â€¢ Verifiable â€¢ <100ms resp â€¢ ZK-Rollups

â€¢ Randomness â€¢ TTL mgmt â€¢ Local fallback

â€¢ ZKP voting â€¢ Hit rate opt â€¢ Proof verification

Solana Batch Processing

â€¢ BatchOracleRequest struct for multiple verifications

â€¢ Single transaction for multiple oracle calls

â€¢ 80% success rate requirement with latency monitoring

â€¢ Event emission for audit trails and performance tracking

Performance Monitoring

â€¢ Sub-second finality tracking (target: <1000ms)

â€¢ Cache hit rate optimization (target: >80%)

â€¢ Oracle uptime monitoring (target: 99.99%)

â€¢ Latency distribution analysis

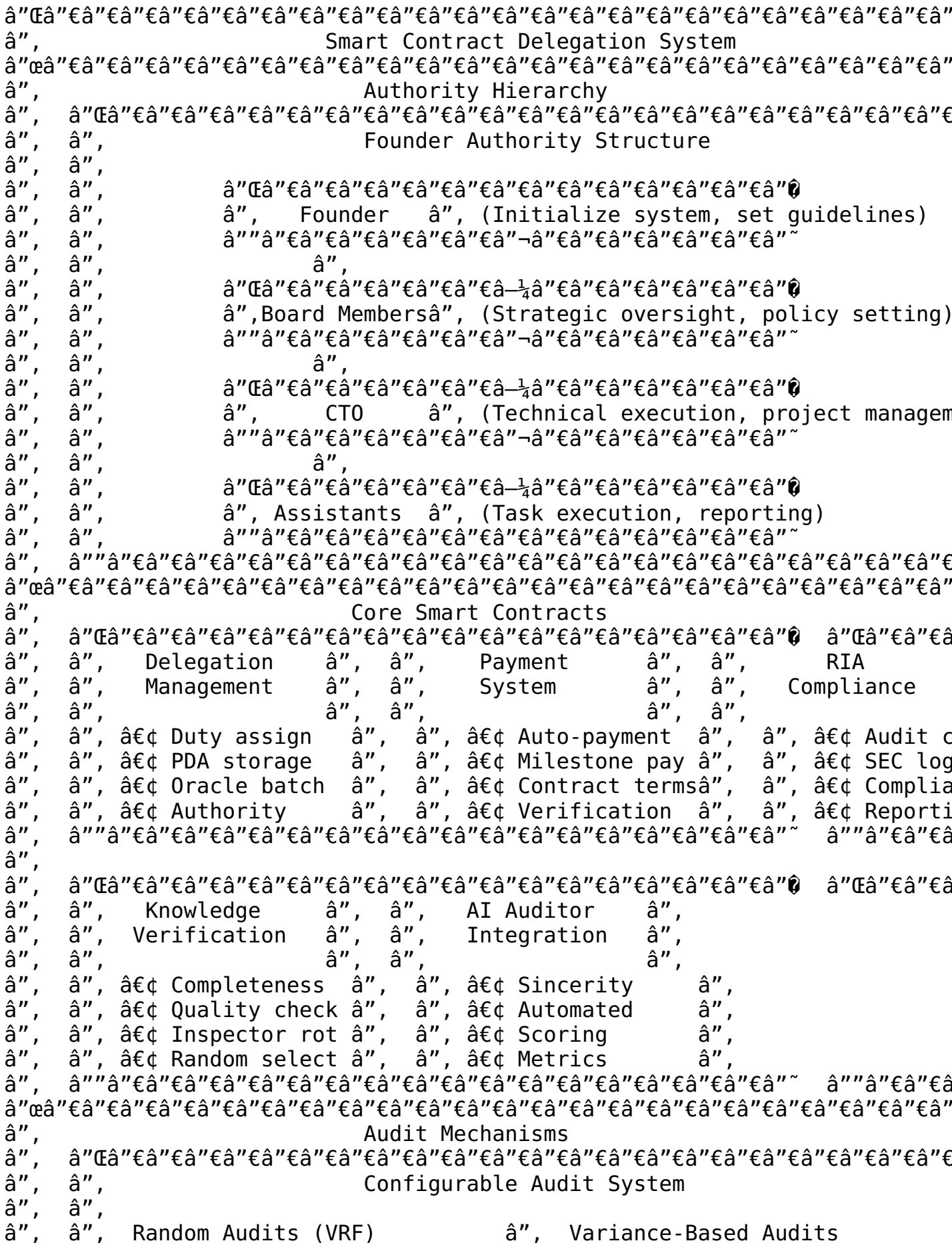
â€¢ Automatic failover and consensus pricing

2. ZKP Voting System Architecture

ZKP Voting Pipeline

	Input Layer					
	User Vote		RL Vote ID		Oracle Data	
	Suggestion		Dynamic Gen		Market	
	Confidence		Causal Links		Verificatio	
	Symbols		ML Enhanced		Sub-sec	
	Processing Layer					
	ZKP Proof Generation					
	Circom		Groth16		Poseidon	
	Circuit		Protocol		Hash	
	Vote anon		Proof gen		Cr	
	Input valid		<2s target		Co	
	Oracle-Enhanced Verification					
	Market data validation through optimized oracles					
	Sub-second verification with cached responses					
	Miden ZK-rollup integration for faster proof verification					
	Chainlink VRF for verifiable randomness in vote selection					
	Storage Layer					
	Neo4j		IPFS		Solana	
	Vote Nodes		Proof Storage		Smart Contract	
	Vote metadata		Immutable		On-chai	
	Relationships		Hash verify		Event e	
	Query index		Distributed		Audit t	
	Security & Compliance					
	Delayed vote detection for manipulation prevention					
	ZKP proof verification with cryptographic guarantees					
	Oracle data integrity through consensus pricing					
	SEC compliance logging with immutable audit trails					
	Performance monitoring for sub-second finality requirements					

3. Smart Contract Delegation Architecture



Throughput Requirements

- **Solana TPS:** 65,000 transactions per second
- **Event Ingestion:** 20,000 events per second
- **Concurrent Users:** 100,000 users
- **Oracle Calls:** 1,000+ per second with batching
- **Database Queries:** 10,000+ per second with caching

Availability Requirements

- **System Uptime:** 99.999% (5.26 minutes downtime/year)
- **Oracle Uptime:** 99.99% with automatic failover
- **Database Availability:** 99.95% with replication
- **API Availability:** 99.99% with load balancing

Security Architecture

Cryptographic Security

- **Quantum-Resistant:** CRYSTALS-Kyber/Dilithium
- **Hash Functions:** SHA-3 for audit trails
- **ZKP Protocols:** Groth16 with Poseidon hash
- **Digital Signatures:** Ed25519 for Solana transactions
- **Encryption:** AES-256-GCM for data at rest

Access Control

- **Multi-Factor Authentication:** Biometric + quantum-resistant
- **Role-Based Access Control:** Hierarchical permissions
- **Smart Contract Permissions:** Authority-based execution
- **API Authentication:** ZKP-based client verification
- **Audit Logging:** Immutable access trail recording

Compliance Architecture

Regulatory Compliance

- **RIA Internet Exception:** Automated compliance with human oversight
- **SEC Rule 10b-5:** Comprehensive disclosure requirements
- **SEC Rule 204-2:** Recordkeeping with 4 hours/year requirement
- **SOC2 Type II:** Automated controls and monitoring
- **GDPR:** Data privacy and right to erasure

Audit Trail Architecture

- **Immutable Logging:** SHA-3 cryptographic hashing
- **Distributed Storage:** IPFS for tamper-proof records
- **Real-time Monitoring:** Prometheus + Datadog integration
- **Compliance Reporting:** Automated SEC-compliant reports

- **Bias Monitoring:** <0.1 threshold for ethical AI compliance
-

Document Version: 1.0

Last Updated: August 2025

Architecture Review: Technical Implementation Guide

Contact: Javier Mandry (@jemandry)